



Experimental and numerical study of formation and development of disturbance waves in annular gas-liquid flow

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ABSTRACT

Disturbance waves in a downwards annular gas-liquid flow were investigated experimentally and numerically in this study. In the experiment, the brightness-based laser-induced fluorescence (BBLIF) technique was utilized to obtain high-resolution spatiotemporal measurements for the film thickness. In the simulations, the two-phase system was simulated by the volume of fluid (VOF) method together with newly developed turbulence damping models, without which the turbulence level around the film surface is considerably under-predicted. Qualitative and quantitative comparisons were carried out for the experimental and numerical data, during which a novel method was developed to extract complex wave structures in a direct manner. Comparisons showed that the model is able to reproduce the main stages of flow evolution, including development of high-frequency initial waves, their coalesce into stable large-scale disturbance waves, generation of slow and fast ripples, and disruption of fast ripples into droplets. The main properties of modeled waves are in decent agreement with the measured ones, apart from noticeably rarer generation of ripples. The presented methods offer a new and promising option to model various energy technology systems, where annular two-phase flow occurs.

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1. Introduction

Annular two-phase flows are ubiquitous in energy technology systems. Yet, the mechanistic behavior of such flows is still not fully understood. In particular, the creation and propagation of disturbance waves, and their role in the mass and energy transfer, still need further investigations. This need is addressed in the present work.

Annular flow is a gas-liquid flow pattern characterized by the presence of liquid film on channel walls and a continuous fast-moving gas core. This pattern occurs at high gas superficial velocities at any flow orientation. Action of turbulent gas stream leads to the appearance of a complex system of waves on a liquid film surface and entrainment of liquid droplets into the gas core. Presence of waves and droplets drastically increases interfacial area and mixing rate, thus enhancing heat and mass transfer in the flow. For

that reason, annular flow is widely used in chemical reactors and cooling systems. Annular flow also occurs in industrial facilities even if it is not organized intentionally. This happens due to separation of gas-liquid mixture with high gas content, e.g., deposition of droplets from a droplet-laden gas flow, condensation of vapor or partial liquid evaporation. Such scenarios are present in boilers, condensers, compact heat exchangers, direct steam generation solar collectors [1], production and transportation pipelines in the oil-and-gas industry [2], in normal operation of boiling water reactors [3] and accident scenarios of pressurized water reactors [4], etc.

Due to its doubtless importance for energy technologies, annular flow has attracted much attention since the late 1950s [5,6] to the present time. Numerous papers are devoted to measurements of the most practically important integral characteristics of the flow – such as pressure drop, entrained fraction, heat transfer coefficient, mean film thickness – and their generalization in the form of empirical correlations. Though this approach may promptly yield useful results, it mostly ignores the flow physics and, as a result, the correlations work well only within the range of conditions in which they were obtained. In order to solve this issue, more

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physics should be taken into account in the corresponding models. Indeed, such integral characteristics of the flow are the result of interaction between gas core and liquid film, which includes formation and development of surface structures of different types and their interaction with the gas stream, leading to even more complex phenomena such as entrainment. The dominant type of surface structures in annular flow is referred to as disturbance waves, which are lumps of liquid traveling with high speed over a thin residual layer of liquid ("base film"). Height of disturbance waves is several times larger than the base film thickness. Disturbance waves coexist with much smaller ripple waves covering the surface of both base film and disturbance waves. From this perspective, accurate modeling of disturbance waves is the key to the reliable prediction of those integral parameters.

Prediction of properties of the waves can also be conducted in an empirical manner, based on experimentally measured values. Many of such correlations are analyzed in a number of recent papers [7–10]. Usage of such relationships and their further incorporation into the correlations for integral flow characteristics leads only to a slightly lower degree of empiricism and the problem of narrow range of applicability does not disappear. A promising approach of predicting disturbance wave behaviors is using the computational fluid dynamics (CFD) method, where wave motions are obtained by solving governing equations locally. Ideally, no empirical correlation is required for the CFD approach, eliminating the application range issue of empirical correlations. However, as a relatively new approach, CFD modeling for disturbance waves is not mature, requiring more effort in model development. At the same time, high-quality experiments are also required to validate corresponding CFD simulations. In the present study, we investigate disturbance waves both experimentally and numerically.

1.1. Suitable experiments for validating CFD results

There are numerous experimental works reporting the average frequency, velocity, amplitude and spacing of disturbance waves. One issue with usage of this kind of data is that the properties of disturbance waves change due to flow development. Study of the effect of downstream distance on properties of disturbance waves were conducted in several papers [11–14]. It is commonly accepted that annular flow gets fully developed at large distances from the inlet. Proper comparison between a model and an experiment can be done by investigating the flow evolution starting from the inlet.

There is no agreement between experimentalists on many important questions, such as the structure of disturbance waves, the importance of ripples, and mechanisms of liquid entrainment. The reason is related to capabilities of measurement techniques employed; many studies were conducted with low-resolution point-wise methods such as conductance probes. Despite that in visual observations the disturbance waves look "milky" [15], in film thickness records obtained by conductance probes they appear as smooth single-crested nonlinear waves with steep front slope [16–18]. This discrepancy was first highlighted by Hewitt et al. [19], who insisted that disturbance waves represent "patches of roughness". In conductance probes records the roughness is smoothed out due to low spatial resolution of the probes. Later, Schubring et al. [20] measured local roughness level on both disturbance waves and base film. It is worth noting that, in the "smooth disturbance waves" studies, temporal records were analyzed, whilst Hewitt et al. [19] and Schubring et al. [20] studied instantaneous spatial records with high spatial resolution.

Alekseenko et al. [21] conducted spatiotemporally resolved measurements with high spatial resolution to study the dynamics of the roughness. It was shown that all the roughness elements, referred to as "ripples", are generated at the rear slopes of

disturbance waves. Two scenarios of further evolution of ripples, namely, "slow" and "fast" ripples, are possible. The slow ripples travel across the base film, and the fast ripples accelerate and travel over the disturbance wave towards its front.

The two concepts of disturbance waves entail different groups of hypotheses describing the mechanism of liquid entrainment from disturbance waves. Within the "smooth" disturbance wave concept, the main hypothesis on mechanism of entrainment is "shearing-off the crests of disturbance waves" [22]. The idea was later developed into quantitative estimations of entrainment intensity [23,24]. Within the "rough" disturbance wave approach, the entrainment occurs due to disruption of ripples on top of disturbance waves, which are shorter and steeper than the disturbance wave in total, and, hence, are more vulnerable to the action of the gas stream. In contrast to "shearing-off the crests", disruption of ripples into droplets via bag- and ligament break-up mechanisms has direct experimental confirmations [25–27].

Another consequence of different approaches is the criterion of identification of disturbance waves. With the "smooth" concept, the criterion can be only quantitative, e.g., the amplitude of a wave should exceed a certain threshold value. With the "rough" approach, the proper qualitative criterion is presence of fast ripples, i.e., generation of short-wavelength perturbations at the rear slope of a wave, which travel across the wave itself with faster speed. As for formation of disturbance waves, there are, again, two main hypotheses: development of long-wave instability of gas-sheared film [5,28] and coalescence of short initial waves [29]. In the former case the wavelength of initial perturbations is assumed to correspond to the distance between disturbance waves, whilst in the latter case the frequency massively decreases when disturbance waves are formed and even before they start to coalesce with each other.

Not many techniques allow for spatiotemporal measurements. Among those which do, multi-component conductance probes [30,31] are of limited spatial resolution. Planar visualization techniques, such as Planar Laser-Induced Fluorescence (PLIF, see Refs. [32–34]) or shadowgraphy approaches [8,10,35] have comparable resolution in radial and axial directions, which makes it rather impossible to significantly expand the region of interrogation in the axial direction. Intensity-based techniques may solve the problem; one of them is Brightness-Based LIF (BBLIF) technique, which will be used in the present work.

1.2. Status of CFD modeling of annular flow

CFD modeling of annular flow is a challenging, complicated and immature process, since two difficult topics, namely multiphase and turbulence, have to be considered simultaneously in the simulations. An overview of the current status of such simulations is given below.

Single-phase simulations were first carried out to investigate disturbance waves where the gas shear effect was modeled as a boundary condition for the interface [36]. In later studies, various two-phase modeling algorithms were adopted. However, except for the Eulerian-Lagrangian approach adopted by Yun et al. [37], the pure Eulerian approach dominated the related studies [38–46], since this approach has better performance in terms of modeling topological changes of the interface. In such examples, either the level-set method [47] or the volume of fluid (VOF) [48] method was always used, since both methods use only one additional equation for the interface advection and treat the two-phase system as a single-phase mixture with a variable density. However, more recent studies [38,39,41,42] favored the VOF method since it inherently conserves mass.

Annular flows with disturbance waves are turbulent, which is

quite challenging for CFD. The so-called direct numerical simulation (DNS) approach is tempting since no turbulence closure is required. However, in order to make such simulations feasible, the periodic boundary condition has to always be employed [44,46], which is unpreferable as discussed in Ref. [38]. Then the large eddy simulation (LES) seems to be a good option since it only needs closures for the subgrid scale (SGS) turbulence, reducing the requirement on computing resources. However, this reduction is rather limited for wall-bounded flows if a wall-resolved LES approach is used [49]. As a result, the wall-modeled LES approach has been adopted in corresponding simulations [39,43]. Since the near-wall region is the key to the thin liquid film modeling, such wall-modeled LES studies are not guaranteed to outperform Reynolds-Averaged Navier-Stokes (RANS) simulations with similar near-wall grid resolutions. Further considering the fact that the gas core is highly turbulent, adopting the RANS approach seems to be an efficient option for CFD simulations on annular flows with disturbance waves.

It should be noted that all SGS and RANS models used in the aforementioned studies are based on widely used single-phase models. Unsurprisingly, directly applying such models to multiphase simulations has its drawbacks, among which the key issue is that the turbulence level around a sheared liquid-gas interface is over-predicted for the gas side and certain corrections are needed to damp the turbulence [50–52]. The Egorov damping approach [51] is a popular model for RANS simulations and is crucial to the accurate modeling of annular flow with disturbance waves [38].

1.3. Goals of the present study

This study is a continuation study of [38] by using newly developed models in the RANS-VOF field [50,53]. This study also uses new experimental data for model validation. In addition, the data processing method is further developed to extract more useful information from the experimental and numerical data. Also, many in-depth discussions are provided on the RANS-VOF modeling and on CFD simulations on annular flow with disturbance waves.

2. Experimental methods

The experiments were carried out in the Kutateladze Institute of Thermophysics. Scheme of the experimental setup and the inlet section is shown in Fig. 1. The working section is a vertical acrylic resin pipe with inner diameter $D = 11.7$ mm and length of 700 mm. Gas enters the working section through a thin-walled metal tube for air supply, mounted coaxially with the main pipe. Liquid enters the working section through a ring-shaped slot between the inner wall of the duct and the outer surface of the gas-supply tube. Thickness of the slot is 0.5 mm, thickness of the wall of the air-feeding tube is 0.3 mm and its inner diameter is 10.1 mm.

Air from the centralized line of compressed air is used as the gas phase. Gas flow rate is measured by an orifice meter connected to a liquid manometer calibrated with a gas flowmeter. The outlet of the working section is open to the atmosphere. Distilled water is used as the working liquid. Prior to the experiments, liquid is cooled down by circulating through the rig in the presence of the gas stream until equilibrium temperature of ~ 16 °C is reached. Fluorescent dye Rhodamine 6G was dissolved in the liquid in small concentration of 10 mg/L, which does not affect physical properties of the liquid. Liquid flow rate is measured by a float rotameter. The water flow is characterized by liquid Reynolds number, Re_L , defined as volumetric liquid flow rate divided by pipe circumference, D , and kinematic viscosity of liquid, ν . In the whole experimental cycle, the range of V_G (gas superficial velocity) is 10–100 m/s and the range of Re_L is 140–400. The effect of gas and liquid flow rates on waves

development will be investigated in a separate paper. In this paper, we restrict the analysis by one combination of flow rates, namely, $Re_L = 350$ and $V_G = 18$ m/s, which is used in the numerical calculations. Pressure drop measurements [54] allow us to estimate the pressure at the inlet for this set of conditions as 1.05 bar.

The thickness of liquid film is measured with BBLIF technique in one longitudinal section of the pipe. In earlier experiments of this kind the length of the region of interrogation (RoI) was about 100 mm [29,55]. In the present experiments, RoI starts from the inlet and its length is 485 mm. This has been done at the expense of lower spatial resolution, which is still high enough to resolve the disturbance waves. For excitation of fluorescence, a longitudinal section of the pipe is illuminated by a vertical laser sheet, emitted by a continuous laser with wavelength of 532 nm and power of 2 W. Thickness of the laser sheet is about 1 mm. RoI is observed with a high-speed camera with a line of 1014 pixels. This gives a spatial resolution of 0.48 mm/pixel. Camera is equipped with a long-pass optical filter to prevent registration of stray laser light. Camera frame rate is 10 kHz and exposure time is half of the period, i.e., 0.05 ms. At each time instance each camera pixel obtains instantaneous fluorescence intensity, (x, t) , in arbitrary camera units. This intensity can be recalculated into local instantaneous film thickness, $h(x, t)$. The interrelation of the two quantities, J and h , are related by equation derived by integrating Lambert-Beer law:

$$J(x, t) = C(x) \left(1 - e^{-ah(x, t)} \right) \left(1 + Ke^{-ah(x, t)} \right), \quad (1)$$

where K is the light reflection coefficient at water-air interface which equals 0.02. Light absorption coefficient a is estimated by measuring the fluorescence intensity values, J_1 and J_2 , emitted by liquid in two slots of known thickness, h_1 and h_2 . The string $C(x)$ is used to convert brightness units into film thickness. This string can be constructed using a reference signal $J_0(x)$, which is intensity of fluorescence emitted by a layer of liquid of known thickness $h_0(x, t)$:

$$C(x) = \frac{J_0(x)}{(1 - e^{-ah_0(x, t)}) (1 + Ke^{-ah_0(x, t)})}. \quad (2)$$

Such a liquid layer can be created by organizing a flow of gravitationally falling film. Though such film is wavy, its time-averaged thickness can be reliably estimated by the Nusselt prediction:

$$h_0 = \sqrt[3]{\frac{3\nu^2 Re_L}{g}}. \quad (3)$$

This prediction works well for Re_L up to 100 and we use the flow with $Re_L = 30$ for the reference signal. Close to the inlet the thickness of the reference film undergoes gradual change from 0.5 mm (inlet slot thickness) to h_0 . This stage is taken into account by introducing dependence $h_0(x)$ at the initial stage on the basis of shadow technique measurements (see Fig. 2(c) in Ref. [35]). More detailed description of BBLIF technique can be found in Ref. [34].

3. Numerical methods

The simulations were carried out at the Royal Institute of Technology using high performance computers. In the following, we focus on key numerical methods used in the present study. Interested readers are referred to Ref. [38] for more details about data processing methods and the numerical setup.

3.1. VOF method

The core concept of the VOF method is to introduce the

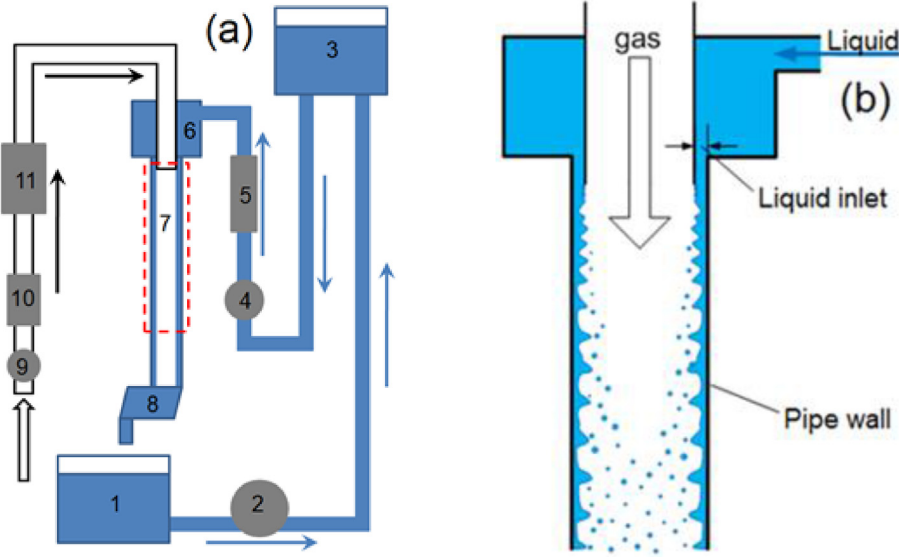


Fig. 1. (a) General scheme of the experimental setup; 1) receiving tank; 2) pump; 3) pressure tank; 4) liquid valve; 5) liquid rotameter; 6) distributor; 7) working section; 8) separator; 9) gas valve; 10) reducer and filter; 11) gas orifice meter. The red dashed rectangle shows the region of interrogation (see also Fig. 2a). (b) Magnified scheme of the inlet section. All the schemes are not to scale. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

volumetric fraction (of the primary phase), α , to model an immiscible two-phase system. Then the two-phase system could be treated as a single-phase system by introducing the mixture density:

$$\rho = \alpha\rho_1 + (1 - \alpha)\rho_2, \quad (4)$$

and the mixture viscosity:

$$\mu = \alpha\mu_1 + (1 - \alpha)\mu_2, \quad (5)$$

where ρ_1 and ρ_2 denote the densities for the primary and the secondary phase, respectively; μ_1 and μ_2 are the dynamic viscosities for the primary and the secondary phase, respectively. In the present study, water and air are set to the primary phase and the secondary phase, respectively. Using these two properties of the mixture, a two-phase system without mass sources could be described by the following equations:

$$\frac{\partial u_i}{\partial x_i} = 0, \quad (6)$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0, \quad (7)$$

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i + f_i^{st}, \quad (8)$$

where τ_{ij} denotes the molecular shear stress tensor; g_i is the gravity vector; $f_i^{st} = -\sigma \frac{\partial n_j}{\partial x_j} \frac{\partial \alpha}{\partial x_i}$ is the volumetric representation of the surface tension force [56] with n_j being the unit normal vector of the interface.

3.2. Turbulence modeling

In OpenFOAM, for isothermal VOF simulations, the following governing equations are used when turbulence models are involved:

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0, \quad (9)$$

$$\frac{\partial \bar{\alpha}}{\partial t} + \tilde{u}_i \frac{\partial \bar{\alpha}}{\partial x_i} = 0, \quad (10)$$

$$\frac{\partial(\bar{\rho} \tilde{u}_i)}{\partial t} + \frac{\partial(\bar{\rho} \tilde{u}_i \tilde{u}_j)}{\partial x_j} = -\frac{\partial \bar{p}^*}{\partial x_i} + \frac{\partial \bar{\tau}_{ij}^{eff}}{\partial x_j} + \bar{\rho} g_i + \bar{f}_i^{st}, \quad (11)$$

where “-” and “~” denote time (ensemble) and Favre averaging, respectively. $\bar{p}^*$ is the modified pressure and $\bar{\tau}_{ij}^{eff}$ is the effective deviatoric stress tensor. For RANS approach, if the Boussinesq eddy viscosity assumption is used, $\bar{\tau}_{ij}^{eff}$ is calculated as:

$$\bar{\tau}_{ij}^{eff} = (\bar{\mu} + \mu_t) \left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} \right), \quad (12)$$

where μ_t is the turbulent dynamic viscosity. Then the goal of corresponding RANS models is to calculate μ_t or $\nu_t = \frac{\mu_t}{\rho}$. In the present study, the following classic k - ω SST model is used:

$$\frac{\partial(\bar{\rho} k)}{\partial t} + \frac{\partial(\bar{\rho} \tilde{u}_i k)}{\partial x_i} = \bar{P} - \bar{\rho} \beta^* \omega k + \frac{\partial}{\partial x_i} \left[(\bar{\mu} + \sigma_k \mu_t) \frac{\partial k}{\partial x_i} \right], \quad (13)$$

$$\begin{aligned} \frac{\partial(\bar{\rho} \omega)}{\partial t} + \frac{\partial(\bar{\rho} \tilde{u}_i \omega)}{\partial x_i} = & \frac{\bar{\rho} \gamma}{\nu_t} P - \bar{\rho} \beta \omega^2 + \frac{\partial}{\partial x_i} \left[(\bar{\mu} + \sigma_\omega \mu_t) \frac{\partial \omega}{\partial x_i} \right] \\ & + 2(1 - F_1) \frac{\bar{\rho} \sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_i} \frac{\partial \omega}{\partial x_i}, \end{aligned} \quad (14)$$

where P symbolizes the production term; F_1 is a blending function; β^* , β , γ , σ_k , σ_ω and $\sigma_{\omega 2}$ are all model coefficients and their detailed values/forms are not discussed in the present study. The following formulation is used to calculate the turbulent viscosity:

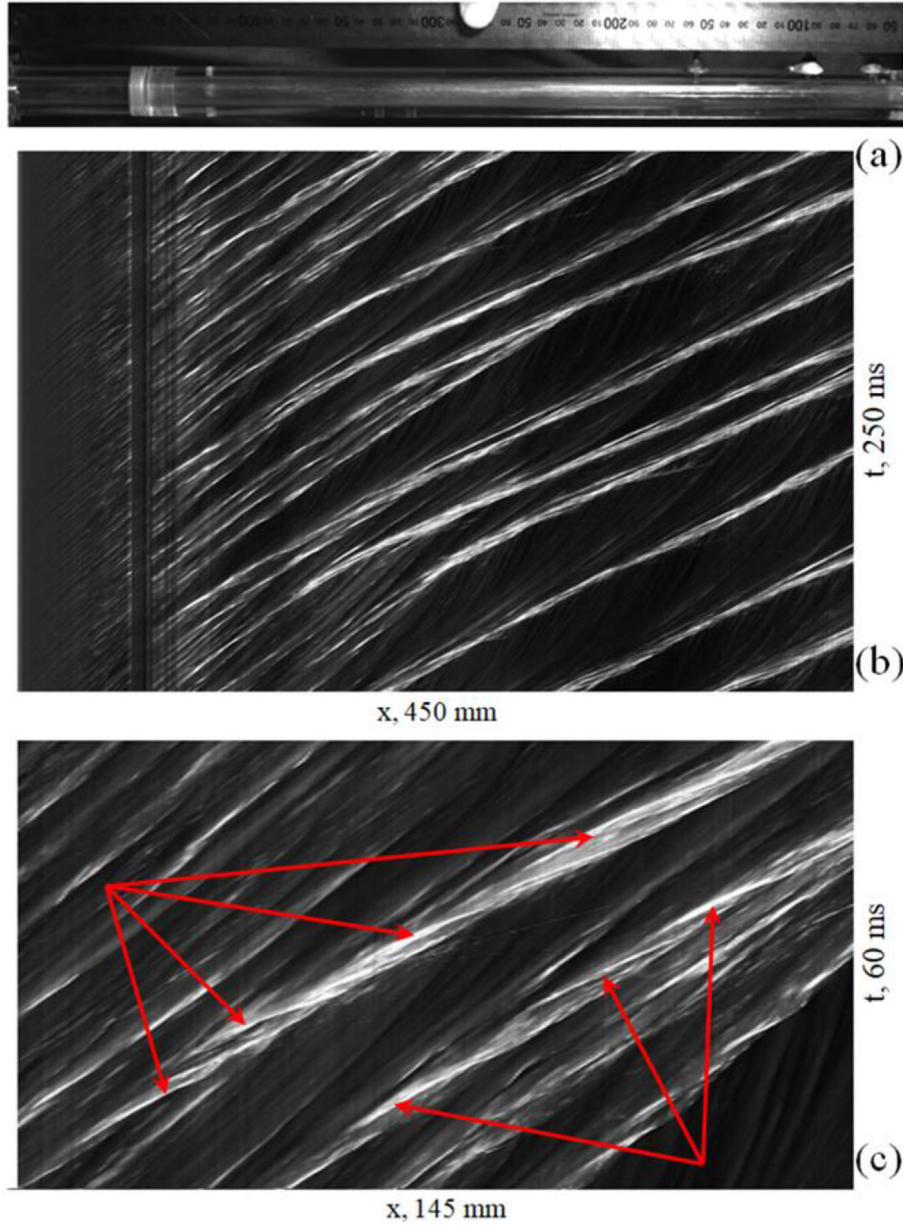


Fig. 2. (a) Image of the pipe with experimental settings of the camera in normal light. The selected fragment is schematically shown by the red dashed rectangle in Fig. 1(a). (b, c) Experimental spatiotemporal matrices of film thickness: full length of the RoI, starting from the inlet (b) and a magnified fragment starting from 180 mm below the inlet with fast ripples marked by arrows (c). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$$\nu_t = \frac{a_1 k}{\max(a_1 \omega, b_1 F_{23} S)}, \quad (15)$$

where a_1 and b_1 are model constants; F_{23} is a blending function; S is the second invariant of the strain rate tensor. The role of the limiter in the denominator is to guarantee the realizability of the model.

3.3. Turbulence damping modeling

In the present study, the turbulence damping effect is phenomenologically modeled by the Egorov approach [51] and its variants [50,57]. The main idea is based on Eq (15), that when the limiter is not activated, i.e., $\nu_t = \frac{k}{\omega}$, a smaller ν_t could be obtained by artificially increasing the value of ω . Two damping approaches,

namely simple and complex, are used in the present study. The simple damping approach adds the following source term to the ω equation:

$$S_{\omega}^{sim} = \text{sign}(B) \frac{36B^2}{\beta \Delta y^3} \left(\frac{A_1 \mu_1^2}{\rho_1} + \frac{A_2 \mu_2^2}{\rho_2} \right), \quad (16)$$

where the damping factor B controls the magnitude of the source term, Δy is the cube root of each individual cell volume. $A_1 = 2\alpha|\nabla\alpha|$ and $A_2 = 2(1-\alpha)|\nabla\alpha|$ are interfacial area densities for water and air, respectively. The $\text{sign}(B)$ term is newly introduced in the present study such that one can introduce a sink term to the ω equation by using a negative B . In such a way, the turbulence could be enhanced.

The complex damping approach adds the following source

terms to Eq. (13). and Eq. (14), respectively [38,53]:

$$S_k^{com} = -\frac{\partial \bar{p}}{\partial x_i} \left[(\bar{v} + \sigma_k v_t) \frac{\partial k}{\partial x_i} \right], \quad (17)$$

$$S_\omega^{com} = -\frac{\partial \bar{p}}{\partial x_i} \left[(\bar{v} + \sigma_\omega v_t) \frac{\partial \omega}{\partial x_i} \right] + \text{sign}(B) \frac{36B^2}{\beta \Delta y^3} \left(\delta \frac{A_1 \mu_1^2}{\rho_1} + \frac{A_2 \mu_2^2}{\rho_2} \right), \quad (18)$$

where δ is a factor to control how the liquid phase is treated. In the original model [51], $\delta = 1$ is used meaning that both phases are symmetrically damped, which is referred to as the symmetric damping approach. While $\delta = -\frac{\rho_1}{\rho_2} \frac{\mu_2^2}{\mu_1^2}$ was proposed in Ref. [50] to model different behaviors of individual phases. This is referred to as the asymmetric damping approach. As shown in Ref. [53], Eq. (17) and Eq. (18) increase the turbulence level considerably with $B = 0$. Therefore, even with a positive B , the net effect of Eq. (17) and Eq. (18) could still be increasing the turbulence level.

4. Results

4.1. Qualitative comparison between experimental and numerical data

Both experimental and numerical results can be presented in the form of spatiotemporal matrices of film thickness, $h(x, t)$, along one longitudinal section of the pipe. A simple and comprehensive way of qualitative analysis of such matrices is creation of their graphical representations, namely, images with local brightness directly proportional to local instantaneous film thickness in the corresponding cell of the matrix. Such examples are given in Fig. 2 for experimental data and in Figs. 3–6 for numerical data. In both cases the white color corresponds to $h = 2$ mm.

4.1.1. Disturbance waves in the experiment

Fig. 2(a) shows an image of the pipe taken by the camera with experimental settings but in white light. The image is turned counterclockwise by 90° . In the experiments, the laser sheet

illuminates the central section of the pipe. In the top (left in the image) part of the pipe the end of the gas-feeding tube can be seen, which corresponds to the beginning of the RoI. About 70 mm downstream the inlet there is a connection between two sections of the pipe. The two sections are connected smoothly on the inside, but the connecting thread blocks optical access to over 10 mm of the pipe length. The effect of the thread is visible in Fig. 2(b).

In the beginning the film surface is covered with high-frequency low-amplitude waves with high degree of regularity (see also [29,35]). Further downstream multiple coalescence actions of these initial waves result in formation of large-scale disturbance waves. The disturbance waves coalesce with each other, which leads to further decrease in waves' frequency. Speed of individual disturbance waves (i.e., the slope of a trajectory with respect to the t axis) increases downstream and reaches saturation in the last 1/3 of the RoI.

The initial waves have nearly no free space between them; the disturbance waves are separated by thin areas of base film covered with slow ripples (see Ref. [21]), generated at the rear slopes of the disturbance waves. They originally move with high speed close to that of the “parent” disturbance waves. But eventually they decelerate and travel across the base film with slow constant speed until the following disturbance wave absorbs them.

The disturbance waves are covered with fast ripples, which do also appear at the rear slopes of the disturbance waves but travel faster than the “parent” disturbance waves. These ripples are marked by arrows in Fig. 2(c), which shows a fragment of $h(x, t)$, matrix in a larger scale. The fast ripples travel across the top of the disturbance wave and disappear either due to decay in front of a disturbance wave, or due to being scattered into droplets by the gas stream.

Thus, in the experiments entrainment occurs due to destruction of fast ripples on top of disturbance waves into droplets by the gas stream. It is not the “crest of disturbance wave” which is broken into droplets; the real disturbance waves do not have “crest”, as it is clearly seen in Fig. 2(c). Their peaks are fast ripples which are constantly renewed; several fast ripples may exist simultaneously on top of a disturbance wave and they always move from back to front of the disturbance wave.

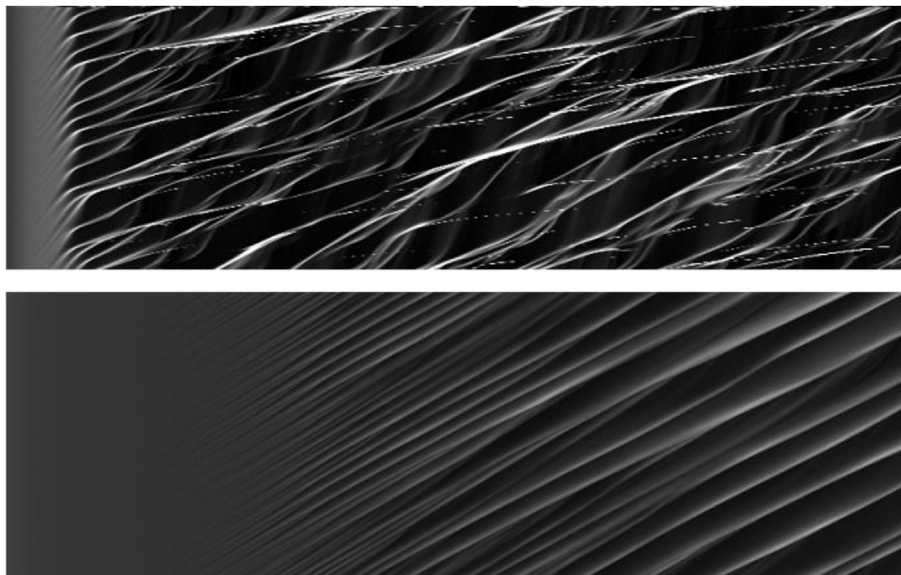


Fig. 3. Numerically computed spatiotemporal matrices with $B = 0$: (a) top: complex damping approach; (b) bottom: simple damping approach. Duration of the records is 250 ms; length of the area is 685 mm.

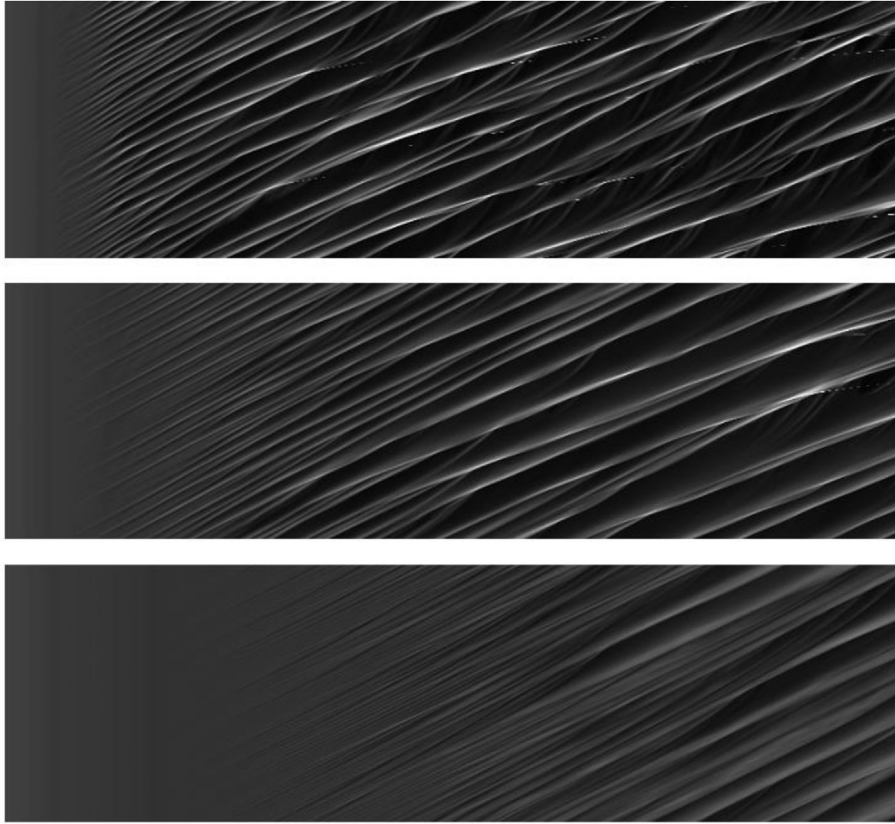


Fig. 4. Numerically computed spatiotemporal matrices with complex approach and symmetric dampening of turbulence: (a) top: $B = 20$; (b) middle: $B = 25$; (c) bottom: $B = 30$. Duration of the records is 250 ms; length of the area is 685 mm.

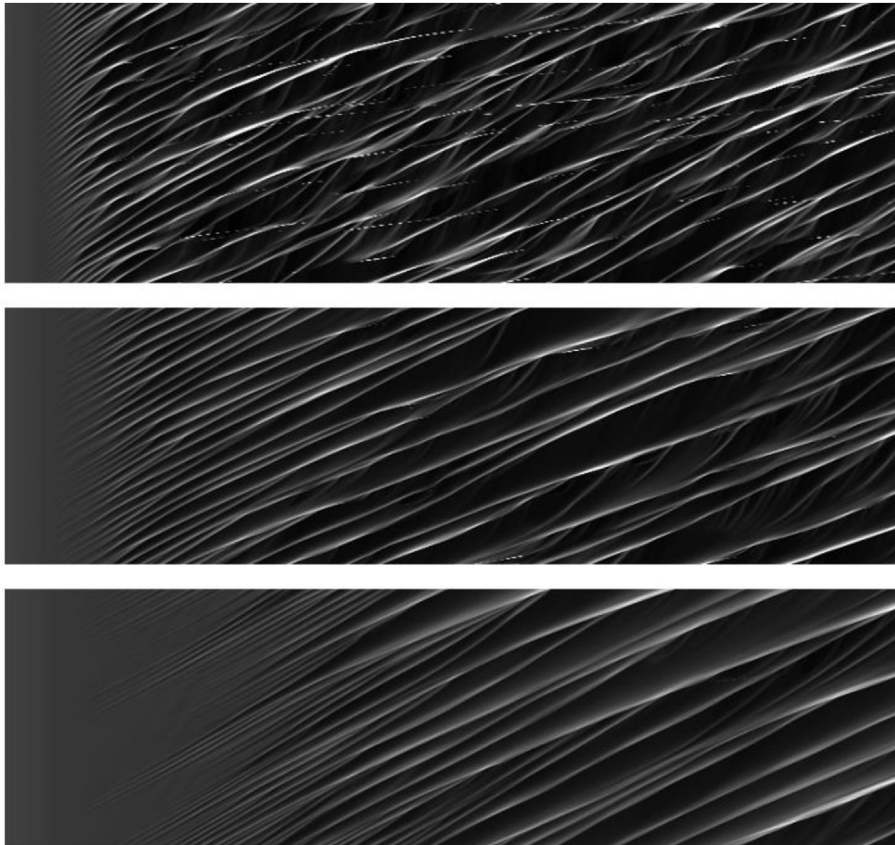


Fig. 5. Numerically computed spatiotemporal matrices with complex approach and asymmetric dampening of turbulence: (a) top: $B = 15$; (b) middle: $B = 25$; (c) bottom: $B = 35$. Duration of the records is 250 ms; length of the area is 685 mm.

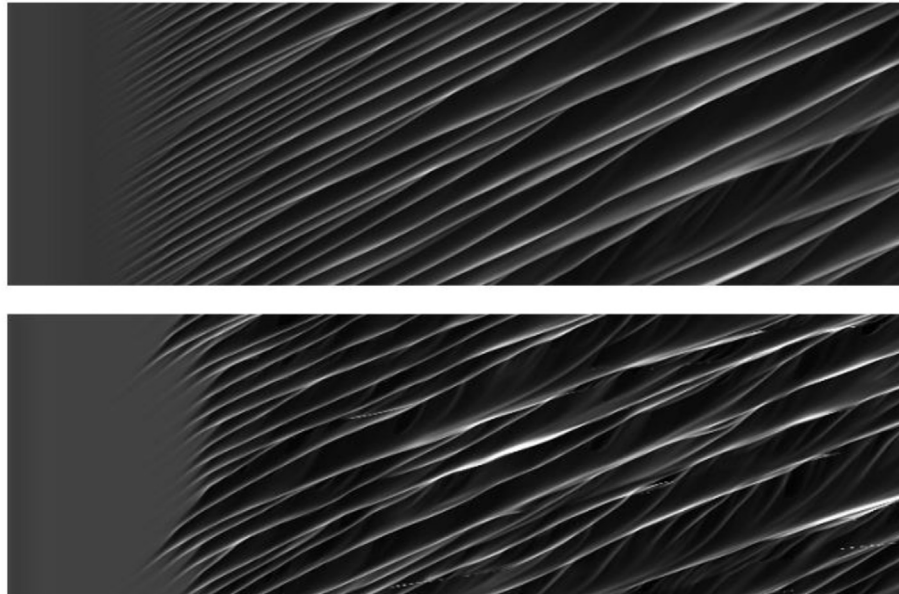


Fig. 6. Numerically computed spatiotemporal matrices with simple approach and turbulence enhancement: (a) $B = -5$; (b) $B = -10$. Duration of the records is 250 ms; length of the area is 685 mm.

Another important feature is the stable existence of a disturbance wave. Once formed, each disturbance wave moves along a stable trajectory (first accelerating, then at a constant speed), and, given it is not absorbed by another disturbance wave, it exists through the whole region of interrogation.

As mentioned in Section 2, the experiments were conducted in a wide range of experimental conditions ($V_G = 10\text{--}100$ m/s and $Re_L = 140\text{--}400$). There is an obvious strong effect of gas and liquid flow rates on wave properties and length of development stages. However, on the qualitative level, the regularities of waves development described above are true for the whole range of conditions.

4.1.2. Numerical results with $B = 0$

The numerically calculated spatiotemporal matrices of film thickness with complex and simple approaches without turbulence modification are shown in Fig. 3. One should note that in the numerical data the spatial size of the matrix is about 1.5 times larger than in the experiments.

The presented matrices have both similarities and differences to the experimental matrix (Fig. 2(b)). At the initial stage in Fig. 3(a) there is also an area with regular high-frequency low-amplitude initial waves, which coalesce to form larger “solitary” waves separated by a thin base film. Generation of slow ripples is also observed. The main difference is that the newly formed waves are single-crested waves with very large amplitude. Their crests can be easily broken into droplets. Such entrainment events occur very frequently in the modeled flow and yield a much larger entrained fraction than in the experiments (see Ref. [38]). When a wave crest is broken, it affects the wave trajectories.

In case of the simple approach (Fig. 3(b)) the stages of flow development are qualitatively the same: regular high-frequency initial waves coalesce to form “solitary” disturbance waves. However, all the stages are much longer than in the experiments: disturbance waves appear only around 500 mm from the inlet. In contrast to complex treatment, the disturbance waves spatiotemporal trajectories are stable, but there are nearly no ripples and no entrainment at all.

4.1.3. Affecting turbulence in the complex and simple approaches

For the complex damping approach, introducing strong ($B \geq 20$) turbulence damping with symmetric treatment suppresses the interfacial instability (Fig. 4). The initial waves appear farther from the inlet and exist for longer distances. Larger number of coalescence events may occur prior to transition to disturbance waves. At $B = 20$ entrainment events are still related to destroying the crest of a wave and still affect the wave’s properties, but to a lesser extent than at $B = 0$. Usually, trajectories of disturbance waves are stable. Generation of slow ripples occurs and has relatively regular character. At $B = 25$ rare and irregular generation of fast ripples is observed; entrainment occurs by destruction of these fast ripples without affecting the disturbance wave itself. Generation of slow ripples is dampened in comparison to $B = 20$. However, at $B = 30$ there is no entrainment within the RoI. At $B = 50$, nearly no waves appear on the film surface.

Increasing B value with asymmetric turbulence treatment (Fig. 5) has qualitatively the same effect on waves as with symmetric treatment. However, quantitatively stronger instability is maintained at the same B values: whilst for symmetric treatment with $B = 30$ no disturbance waves are formed in the region, they are present at $B = 35$ for asymmetric treatment. It is not surprising, since turbulence level even increases inside the liquid phase near the surface. The entrainment often occurs via destruction of fast ripples, as in the experiments, though production of fast ripples is much less intensive than in the experiments.

Applying negative B within the simple approach enhances instability, which leads to earlier development of waves, generation of slow ripples, fast ripples and occurrence of entrainment (Fig. 6). In general, this picture is similar to that of the complex approach with moderate B values. This similarity indicates that the net effect of Eq (17), and Eq (18), is increasing the turbulence level even though positive B values are used. These all suggest that the turbulence level is under-predicted around the film surface if the single-phase turbulence model is used directly. This phenomenon is different from previous studies where interfacial turbulence was over-predicted [50,51,58]. The reason is that previous simulations were conducted with interfaces relatively far away from the wall such that there is limited or no wall effect. However, the wall effect

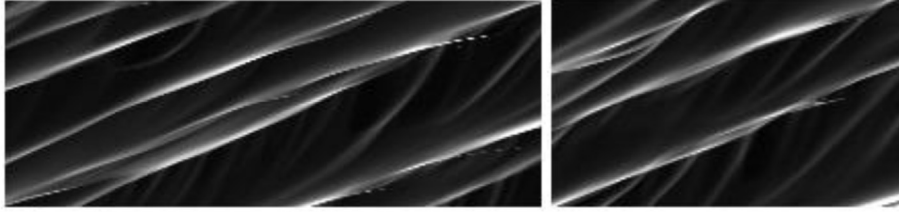


Fig. 7. Magnified examples with fast ripples observed in the model data. (a) left: complex approach, asymmetric treatment, $B = 25$. The fragment size is 220 mm by 100 ms. (b) right: simple approach, $B = -10$. The fragment size is 145 mm by 100 ms.

cannot be neglected for film surfaces in annular flows [59]. This is an interesting phenomenon that requires more in-depth studies.

Magnified examples of fast ripples generation and disruption into droplets are shown in Fig. 7. They show similar behavior to that of experimentally observed fast ripples, being produced at the rear

end of a disturbance wave and disappearing at its front, sometimes leaving a track of entrained droplet. The difference to the experiments is that the frequency of appearance of fast ripples is much lower in the model. Majority of modeled disturbance waves remains single-crested even for the “best” treatment cases.

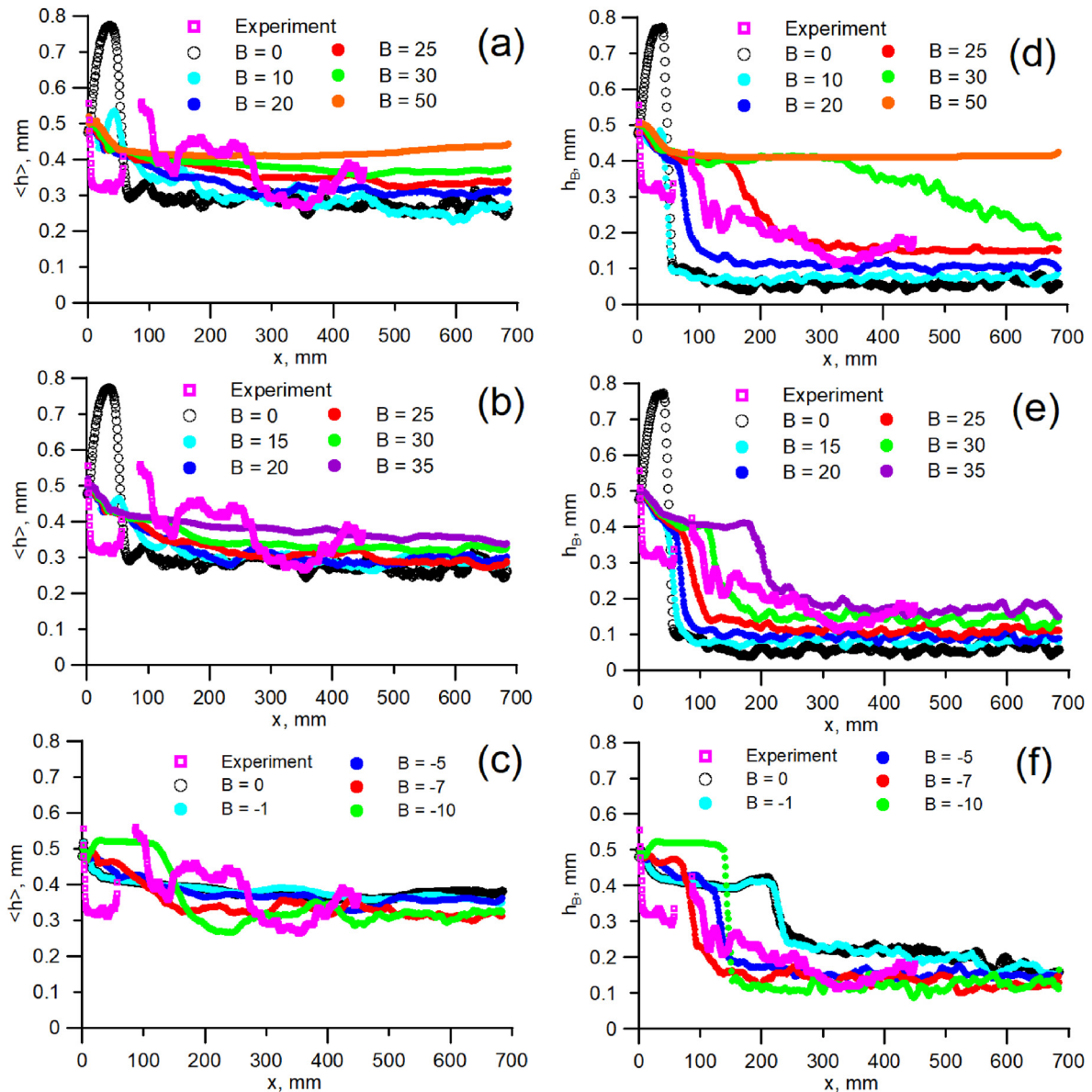


Fig. 8. Comparison of downstream evolution of mean (a, b, c) and modal (d, e, f) film thickness between the experiments and model for complex damping with symmetric (a, d) and asymmetric (b, e) turbulence treatment and for simple damping (c, f).

4.2. Quantitative comparison between experimental and numerical data

In this section, downstream evolution of the main properties of liquid film and waves will be compared quantitatively between the experiments and different modeling approaches. Mean film thickness, $h(x)$, decreases gradually with downstream distance from initial 0.5 mm (liquid outlet slot thickness) to about 0.3–0.4 mm for nearly all the models (Fig. 8a–c). Stronger turbulence slightly reduces the mean film thickness. Exception is the complex approach with $B = 0$: a thick rim appears immediately as the liquid leaves the distribution slot. The rim does not appear in any other model.

The modal value is an estimate of the base film thickness, $h_B(x)$. In contrast to h , h_B undergoes sudden decrease as soon as the initial waves start to grow and coalesce (Fig. 8d–f). After that, h_B is stabilized promptly on a certain low level. The higher is the level of turbulence, the closer to the inlet the transition to thin base film occurs and base film gets noticeably thinner. No transition is observed for weak turbulence ($B = 50$, symmetric treatment).

Properties of the waves are usually measured with either

indirect or direct approaches [60]. The former implies extraction of characteristics of a whole film thickness record and attributing the obtained characteristics to the properties of waves of a particular type. Direct methods rely on identification of each individual wave of a particular type in a record. Identification can be done either on amplitude-based criterion [8,14,61], or combination of amplitude and velocity [29,55,60]. In the present paper, we adapt the algorithm in Ref. [62] to flow with entrainment.

First, local maxima in both temporal and spatial records of film thickness are found. Then the marked maxima are connected into characteristic lines. Too short lines are discarded as erroneous. Further, separation of found waves into two types is conducted based on a threshold value of product of peak height and velocity of a wave, $h_p V$. Examples of found characteristic lines for waves of two types are shown in Fig. 9. The red lines correspond to “fast” waves (which are first the initial waves and then the disturbance waves and, if they exist, fast ripples) and green lines denote trajectories of the slow ripples. Note that the spatial and temporal scales are different for modeled and experimental results. The most prominent difference between the numerical and experimental data is

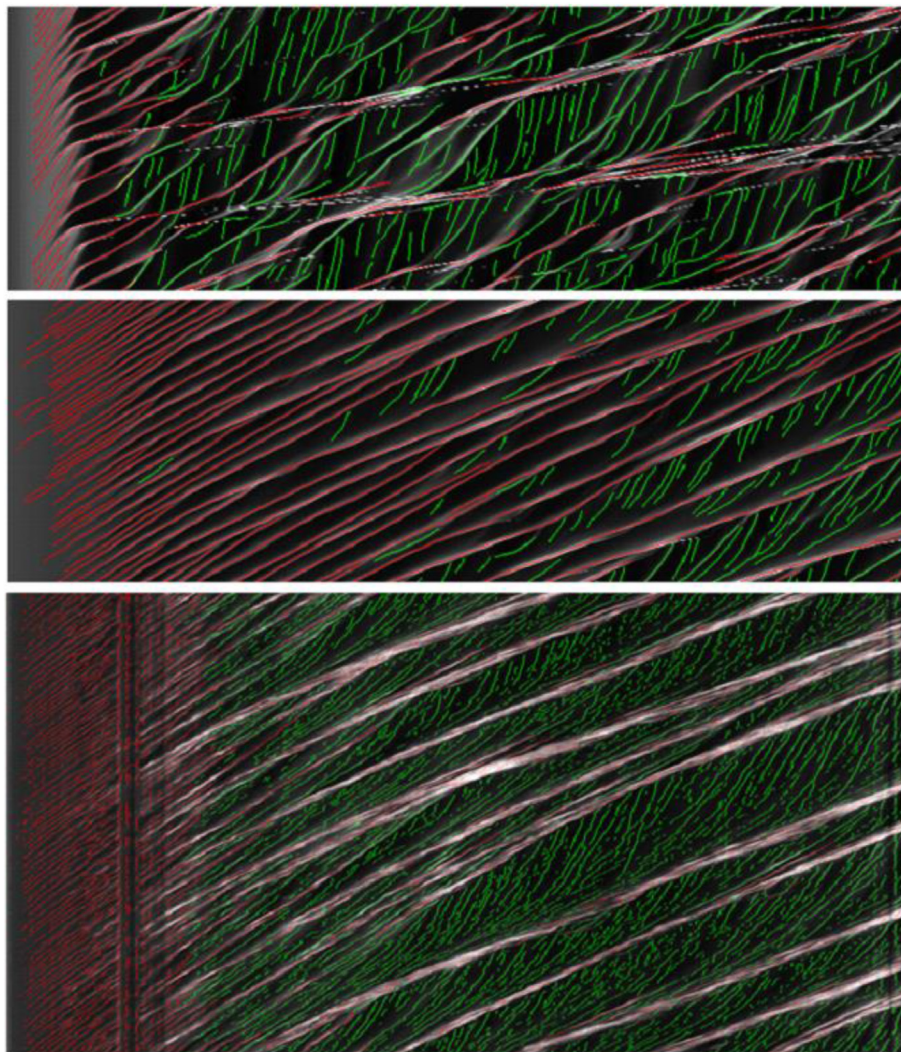


Fig. 9. Visualization of algorithm application to identify initial waves, disturbance waves and fast ripples (red lines) and slow ripples (green lines). (a) top: complex approach, $B = 0$. (b) middle: complex approach, asymmetric treatment, $B = 25$. Temporal duration of each fragment is 0.25 s, spatial length 685 mm. (c) bottom: identification of the characteristic lines in the experimental data. Temporal duration is 0.25 s and spatial length is 480 mm. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

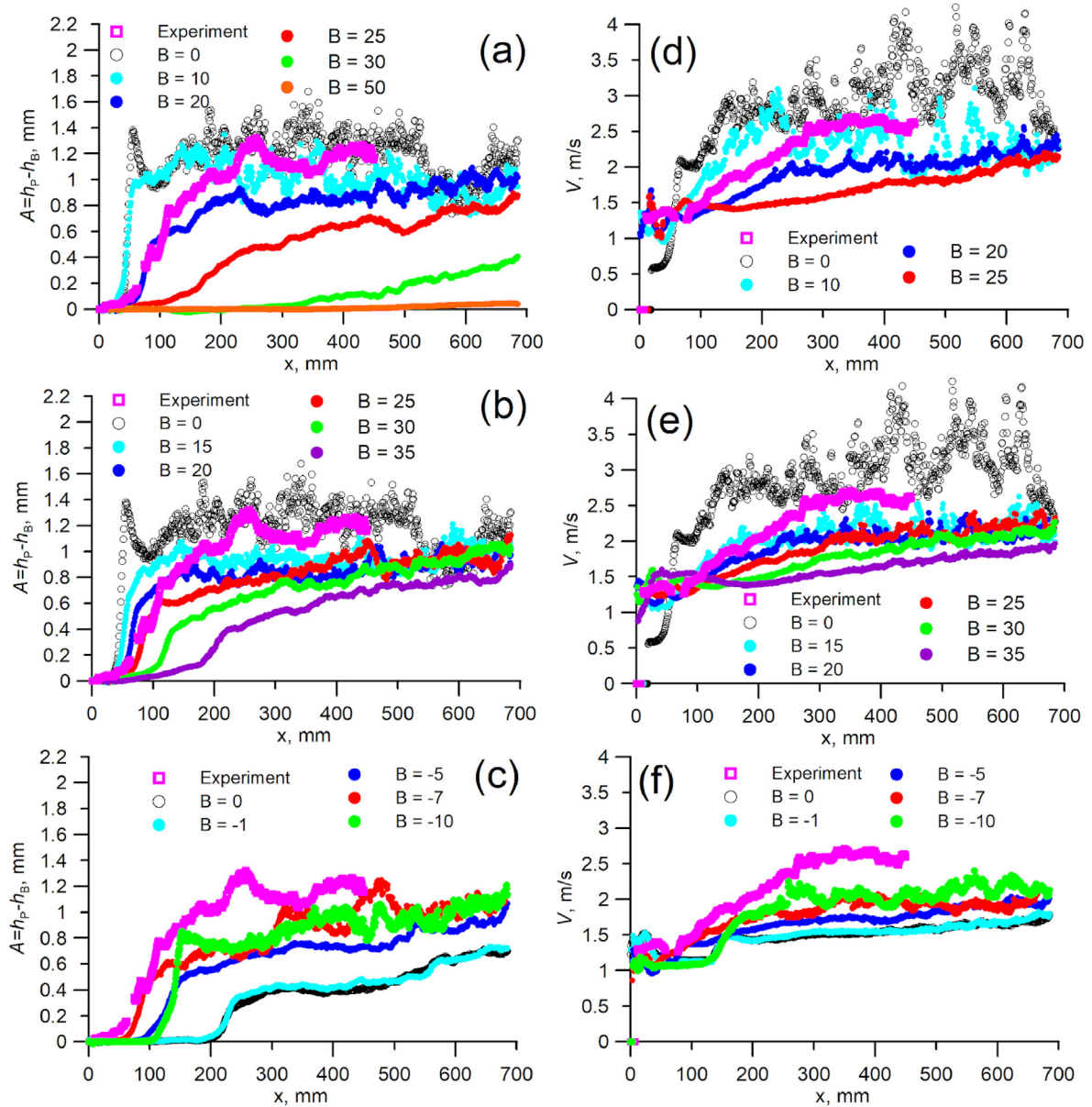


Fig. 10. Comparison of downstream evolution of amplitude (a, b, c) and velocity (d, e, f) of “fast” waves between the experiments and model for complex damping with symmetric (a, d) and asymmetric (b, e) turbulence treatment and for simple damping (c, f).

the much larger number of fast and slow ripples in the experiments.

Based on obtained characteristic lines, main properties of waves can be measured at each downstream position and compared between experiments and different models, as shown in Figs. 10–11.

The *amplitude* of the fast waves, A , is defined as difference between film thickness at the peak of a wave, h_p , and the base film thickness, h_b . Fig. 8(a–c) shows the mean value of over all the waves detected at a certain x . Downstream coordinate of the sharp decrease in base film thickness corresponds well to the sharp increase in the waves’ amplitude, as shown in Fig. 6(d–f) and Fig. 8(a–c). The best correspondence to the experiments is observed for $B = 20$ for both kinds of treatment, though for the asymmetric treatment both $B = 15$ and $B = 25$ are not too different. The experimental saturated level of amplitude is close to the high-turbulence numerical conditions, but about 1.5 times higher than the moderate-turbulence case. On the other hand, BBLIF may yield

overestimation of fast ripples amplitude at steep interfacial slopes [55].

The *velocity* of the fast waves, V , is defined as the local slope of a characteristic line with respect to the t -axis. V shows quite stable values in the range of 1–1.5 m/s within the initial wave region (Fig. 8 d–f). In the experiments, the velocity of the initial waves corresponds well to that in the model. Similar to the amplitude of the waves, the saturated velocity value is close to that of modeled waves under strong turbulence and exceeds that of moderate turbulence. It should be borne in mind that the measured velocity value describes the fast ripples, which move faster than the disturbance waves. In the model, the appearance of fast ripples is a rare event. The velocity growth rate is closer to the moderate turbulence case.

The *frequency* of the fast waves, f , is defined as the number of characteristic lines crossing the column of the $h(x, t)$ matrix with coordinate x ; in general f decreases with the downstream distance

due to coalescences of initial waves and disturbance waves (Fig. 11a–c). Qualitatively, the frequency behavior in the flow direction is similar in the model and the experiments; the difference is mainly in much more abrupt frequency change at the transition. The second experimental plot shows an estimation of the frequency of disturbance waves as the number of waves separated by large enough time spans. Quantitatively, the disturbance wave frequency is close to that of high-turbulence numerical data, but about 1.5 times lower than the moderate-turbulence results.

The longitudinal size of the fast waves, L , is defined as the distance between the points at the front and rear slopes of a wave, where film thickness drops below a certain threshold. The threshold value is set to $h_B + 0.2(h_P - h_B)$. L is of order of several mm for the initial waves (Fig. 11d–f). Due to the coalescence of the initial waves, not only the amplitude, but also the longitudinal size increases. It shows similar behavior with x to that of the amplitude.

To summarize, the main properties of the waves show the same

behavior with downstream distance as in the experiments and the values of the wave parameters at different stages of evolution are also in decent agreement. The main discrepancies are related to frequencies of small-scale waves: whilst frequency of disturbance waves is overestimated by the model, frequency of initial waves and frequency of generation of slow and especially fast ripples is substantially underestimated. We consider that one of the major challenges in future modeling efforts is how to enhance this short-wave instability while maintaining the large-scale disturbance waves stable over a long distance.

4.3. Further discussions based on CFD results

In comparison with the experiments, the most outstanding feature of CFD simulations is that detailed information is available for the entire computational domain. This allows us to get more insights into the complicated phenomena, which are still not yet

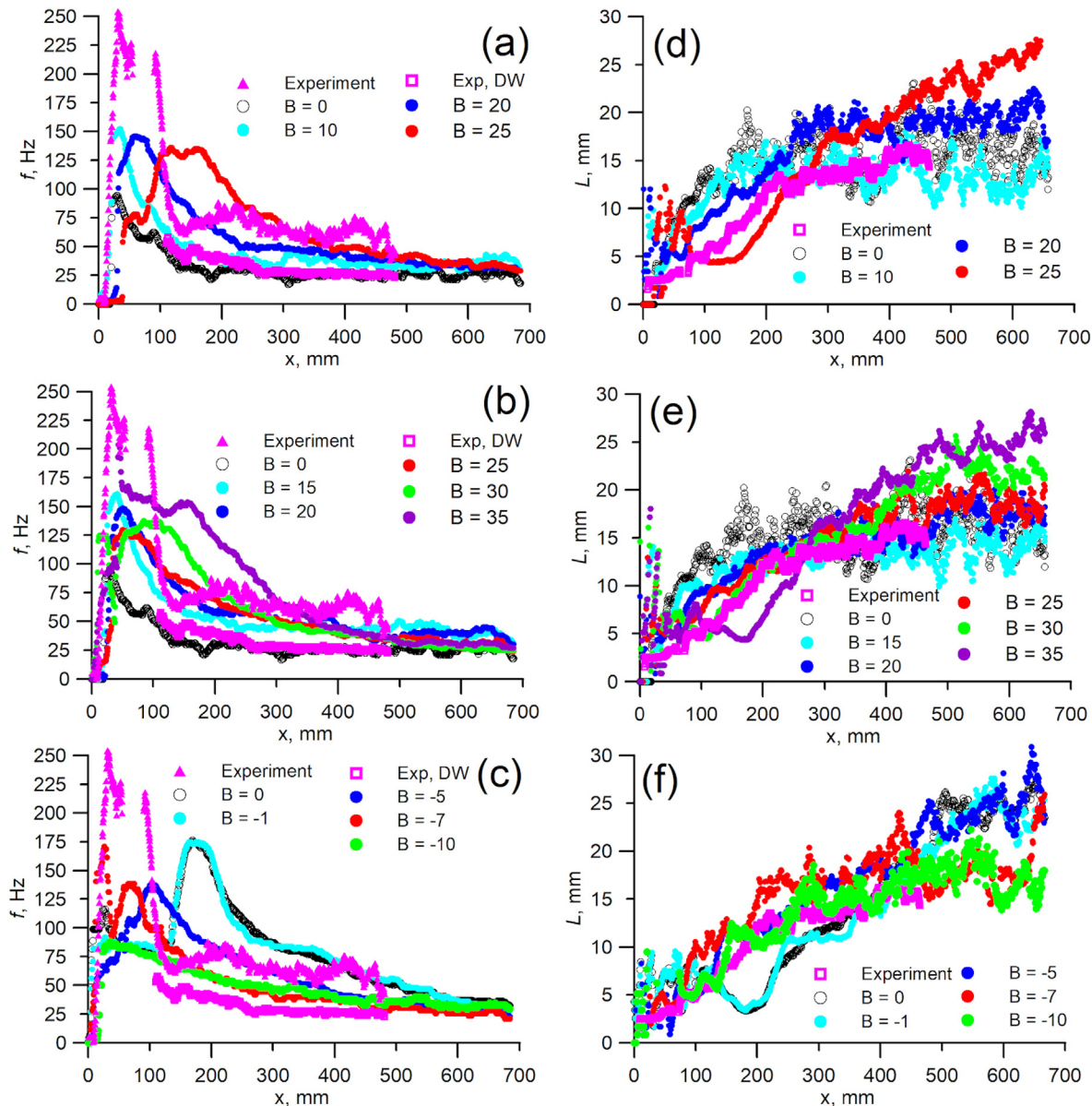


Fig. 11. Comparison of downstream evolution of frequency (a, b, c) and longitudinal size (d, e, f) of “fast” waves between the experiments and model for complex damping with symmetric (a, d) and asymmetric (b, e) turbulence treatment and for simple damping (c, f).

available based on the current measurement techniques. In the following, we show instantaneous and localized distributions of several important parameter in the (θ, x) coordinate system, where θ is the polar angle and x is the distance from the inlet.

4.3.1. Film thickness

All the aforementioned comparisons are based on the film thickness data along a sampling line in the flow direction, i.e., a given θ , due to the limitation of the experiment. In the simulations, the film thickness distribution is available for the whole domain, and an instantaneous film distribution is given in Fig. 12. Two key features of disturbance waves, namely, high amplitude and large wavelength, are intuitively visualized in this picture. In addition, disturbance waves maintain a certain level of uniformity in the circumferential direction, which is consistent with the experimental observation [14].

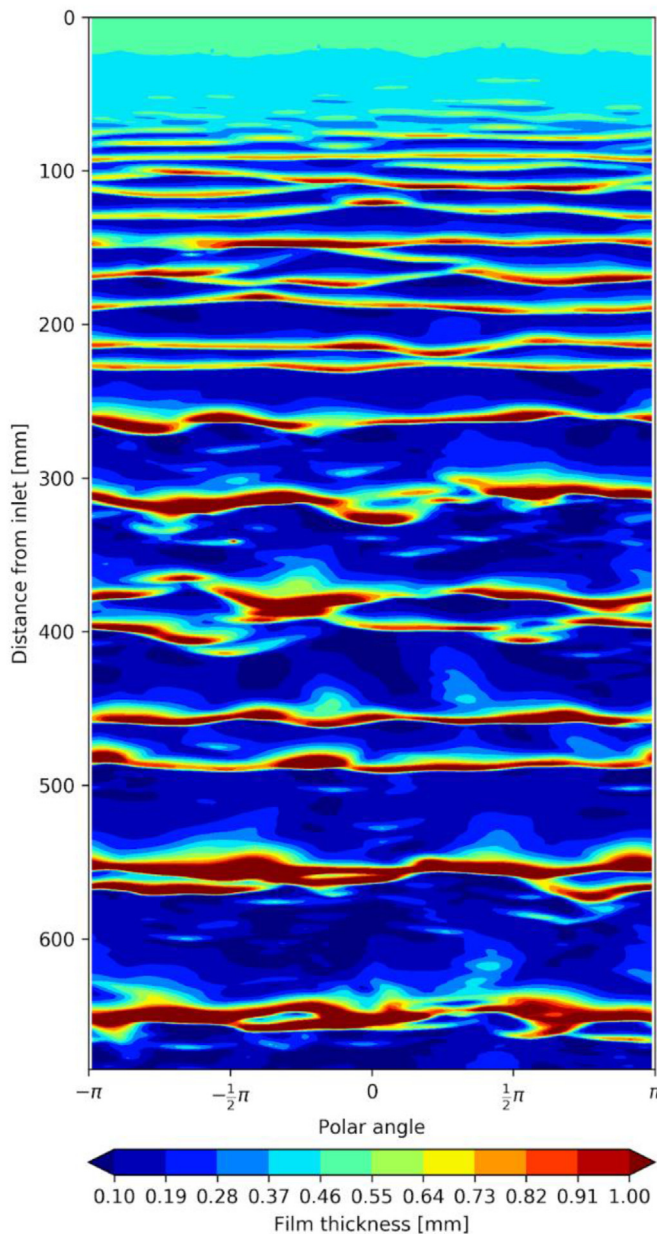


Fig. 12. Instantaneous film thickness distribution.

4.3.2. Wall shear stress

With access to the velocity field, the wall shear stress is calculated. In Fig. 13, the instantaneous axial wall shear stress corresponding to Fig. 12 is provided. Since disturbance waves have a much larger speed than the base film, the wall shear stress peaks at the disturbance wave regions. Besides, there is a clear distinction between the upstream and downstream of disturbance waves. The rear slopes have intermediate stresses, while regions just in front of disturbance waves have the smallest, even occasionally negative, shear stresses. Therefore, flow reversal may happen in wave fronts.

4.3.3. Interfacial shear stress

Interfacial shear stress plays a key role in the propagation of waves. Even though the effective deviatoric tensor is defined and calculated by Eq. (12), its result for the interfacial region is more numerical than physical due to the blending of material properties. Therefore, we can only conduct a qualitative study based on the

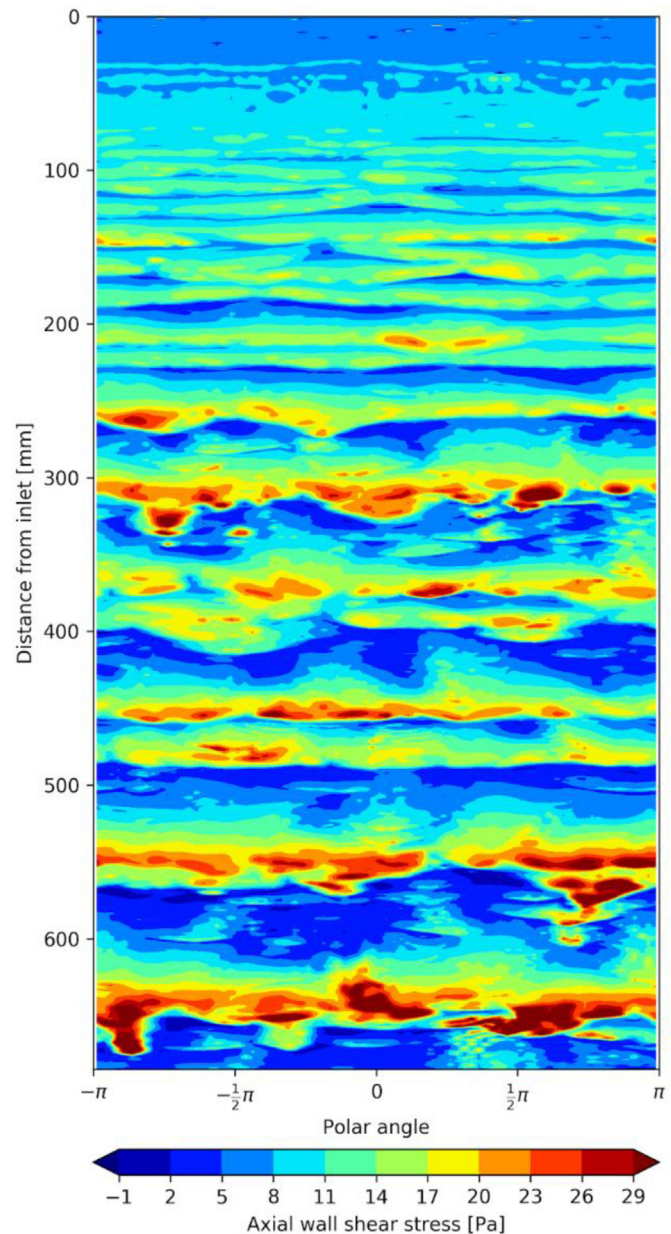


Fig. 13. Instantaneous axial wall shear stress.

magnitude of $\bar{\tau}_{ij}^{eff}$. We further normalize it by the mean value of the entire interface, as shown in Fig. 14. The distribution is similar to the wall shear stress that it peaks at disturbance waves and reduces to a rather low level at wave fronts. It is this large shear stress that sustains the fast moving of disturbance waves.

4.3.4. y^+ for the interface

$y^+ = \frac{u_\tau y}{\nu}$ is an important parameter in the modeling of near-wall flows, where u_τ is the friction velocity and y is the physical distance to the solid wall. If we evaluate y^+ at the film surface, we get another form of the liquid film Reynolds number, y_i^+ , which was used as an entrainment criterion [22,63–65]. The largest value recommended by those references for y_i^+ is 18. The result for y_i^+ is shown in Fig. 15, where the majority of the base film has a y_i^+ value smaller than 30, meaning that the base film is so thin that it resides

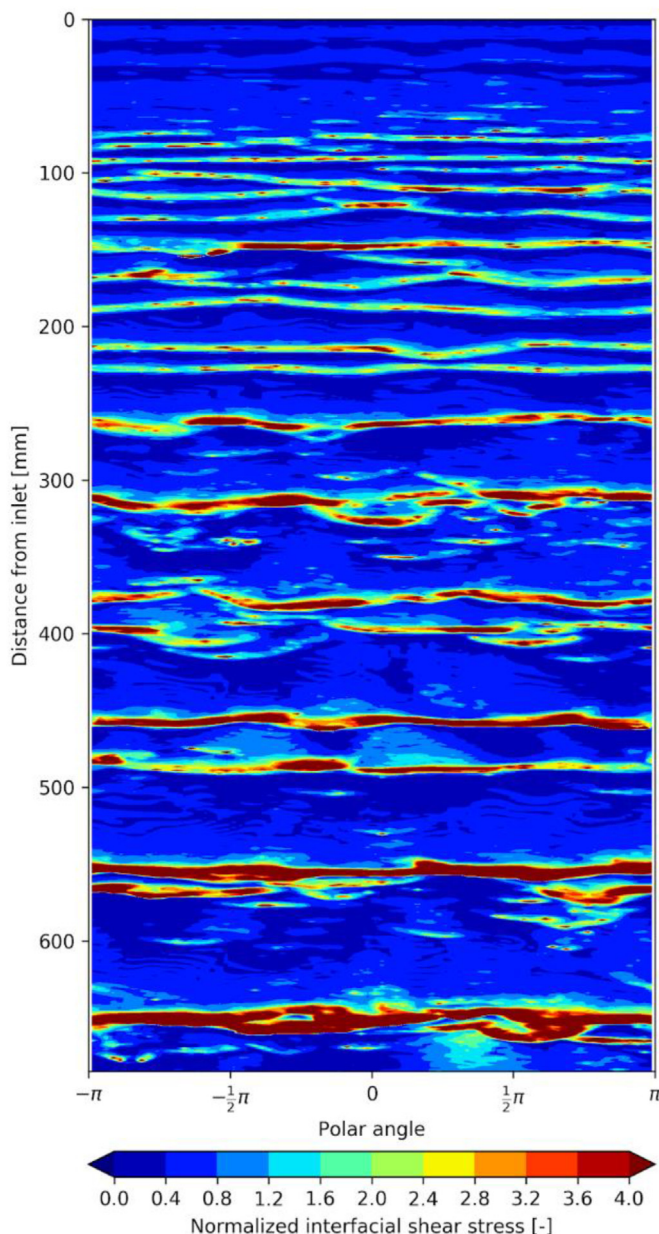


Fig. 14. Normalized interfacial shear stress.

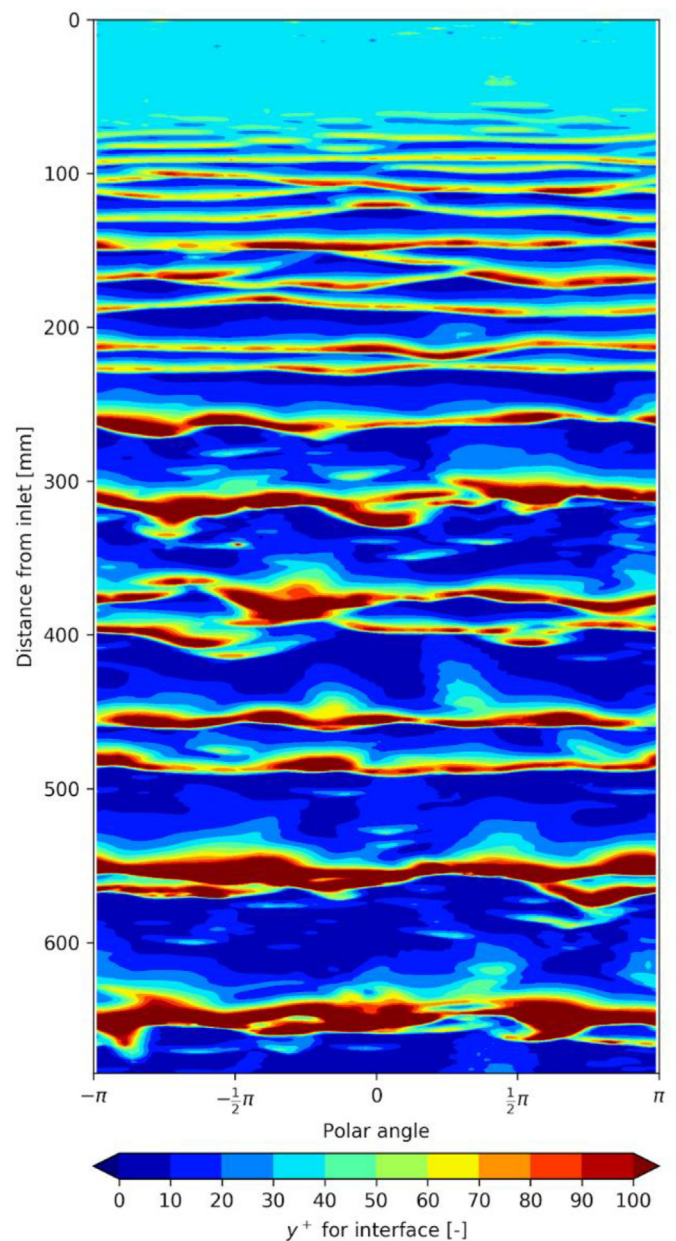


Fig. 15. y^+ for the interface.

in the buffer region and the viscous sublayer of the boundary layer, justifying the requirement of a fine near-wall mesh in such CFD simulations. y_i^+ of disturbance waves can go beyond 100 but mainly reside in the log region of the boundary layer.

5. Conclusions and outlooks

Experiments and simulations were carried out to investigate the formation and development of disturbance waves in a downwards air-water annular flow. In both the experiment and simulations instantaneous profiles of film thickness in the whole region of interrogation were obtained with high sampling frequency and spatial resolution. The region of interrogation is a long (0.5–0.7 m) section of the pipe, which starts from the inlet.

Experimental study reveals the main stages of waves' evolution with downstream distance: generation and development of high-

frequency quasi-regular initial waves, coalescence of initial waves into large-scale disturbance waves, acceleration and coalescence of disturbance waves with further stable propagation. The disturbance waves generate slow and fast ripples on their rear slopes; disruption of fast ripples into droplets leads to entrainment.

Various modeling approaches were applied to reproduce the main stages of evolution observed in the experiments and some of them were successful, on both qualitative and quantitative levels. Downstream change of mean and base film thickness, along with amplitude, velocity, frequency and longitudinal size of the waves show the same behavior with distance, similar distances at which certain transitions occur and close values at the corresponding stages of evolution. The main difference is lower frequency of small-scale waves (initial waves, slow and fast ripples).

Comparisons between simulations and experiments reveal a unique issue with RANS-VOF simulations of annular flows that, when the single-phase turbulence model is used without corrections, the turbulence level around the interface in annular flow is significantly under-predicted. This issue was resolved in a phenomenological manner; the turbulence treatment was selected so that the model would reproduce the main phenomena in wave formation and development processes. Such phenomena are qualitatively the same in a wide range of gas and liquid flow rates; thus, we expect the selected turbulence treatment would work in other conditions as well. However, mechanistic models should be further developed to make accurate and reliable predictions for annular flows in energy technology systems.

Apart from future development of the model, further steps of model validation should be done in future, namely: a) Validation of the model in a wider range of gas and liquid flow rates; b) Validation of global properties of the flow – such as shear stress or entrainment rate – extracted from the modeled data; c) Validation of instantaneous velocity fields in both phases obtained from the modeled data; d) Validation of three-dimensional wave structure predicted by the model; e) Validation of the model in different flow configurations such as changing shape, size and orientation of the duct, properties of working fluids, etc.

All the procedures above require intensive numerical and/or experimental efforts and cannot be done within the frame of one paper. However, these procedures are promising for prediction of annular flow parameters for various industrial applications.

CRedit authorship contribution statement

Wenyuan Fan: Conceptualization, Funding acquisition, Methodology, Software, Visualization, Data curation, Writing - original draft. **Andrey V. Cherdantsev:** Formal analysis, Funding acquisition, Investigation, Methodology, Visualization, Writing - original draft. **Henryk Anglart:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Sandá A, Moya SL, Valenzuela L. Modelling and simulation tools for direct steam generation in parabolic-trough solar collectors: a review. *Renew Sustain Energy Rev* 2019;113:109226. <https://doi.org/10.1016/j.rser.2019.06.033>.
- [2] van Nimwegen AT, Portela LM, Henkes RAWM. Modelling of upwards gas-liquid annular and churn flow with surfactants in vertical pipes. *Int J Multiphas Flow* 2018;105:1–14. <https://doi.org/10.1016/j.jmultiphaseflow.2017.09.012>.
- [3] Le Corre JM. Experimental characterization of boiling two-phase flow structures under BWR core operating conditions. *Nucl Eng Des* 2020;365:110708. <https://doi.org/10.1016/j.nucengdes.2020.110708>.
- [4] Lokanathan M, Hibiki T. Flow regime transition criteria for co-current downward two-phase flow. *Prog Nucl Energy* 2018;103:165–75. <https://doi.org/10.1016/j.pnucene.2017.11.021>.
- [5] Hanratty TJ, Engen JM. Interaction between a turbulent air stream and a moving water surface. *AIChE J* 1957;3:299–304. <https://doi.org/10.1002/aic.690030303>.
- [6] van Rossum JJ. Experimental investigation of horizontal liquid films. *Chem Eng Sci* 1959;11:35–52. [https://doi.org/10.1016/0009-2509\(59\)80071-3](https://doi.org/10.1016/0009-2509(59)80071-3).
- [7] Berna C, Escrivá A, Muñoz-Cobo JL, Herranz LE. Review of droplet entrainment in annular flow: interfacial waves and onset of entrainment. *Prog Nucl Energy* 2014;74:14–43. <https://doi.org/10.1016/j.pnucene.2014.01.018>.
- [8] Pan L, He H, Ju P, Hibiki T, Ishii M. Experimental study and modeling of disturbance wave height of vertical annular flow. *Int J Heat Mass Tran* 2015;89:165–75. <https://doi.org/10.1016/j.ijheatmasstransfer.2015.05.073>.
- [9] Dasgupta A, Chandraker DK, Kshirasagar S, Reddy BR, Rajalakshmi R, Nayak AK, et al. Experimental investigation on dominant waves in upward air-water two-phase flow in churn and annular regime. *Exp Therm Fluid Sci* 2017;81:147–63. <https://doi.org/10.1016/j.expthermflusci.2016.10.012>.
- [10] Lin R, Wang K, Liu L, Zhang Y, Dong S. Application of the image analysis on the investigation of disturbance waves in vertical upward annular two-phase flow. *Exp Therm Fluid Sci* 2020;114:110062. <https://doi.org/10.1016/j.expthermflusci.2020.110062>.
- [11] Hall Taylor NS, Nedderman RM. The coalescence of disturbance waves in annular two phase flow. *Chem Eng Sci* 1968;23:551–64. [https://doi.org/10.1016/0009-2509\(68\)89003-7](https://doi.org/10.1016/0009-2509(68)89003-7).
- [12] Wolf A, Jayanti S, Hewitt GF. Flow development in vertical annular flow. *Chem Eng Sci* 2001;56:3221–35. [https://doi.org/10.1016/S0009-2509\(00\)00546-7](https://doi.org/10.1016/S0009-2509(00)00546-7).
- [13] Hazuku T, Takamasa T, Matsumoto Y. Experimental study on axial development of liquid film in vertical upward annular two-phase flow. *Int J Multiphas Flow* 2008;34:111–27. <https://doi.org/10.1016/j.jmultiphaseflow.2007.10.008>.
- [14] Zhao Y, Markides CN, Matar OK, Hewitt GF. Disturbance wave development in two-phase gas–liquid upwards vertical annular flow. *Int J Multiphas Flow* 2013;55:111–29. <https://doi.org/10.1016/j.jmultiphaseflow.2013.04.001>.
- [15] Hewitt GF, Hall Taylor NS. Annular two-phase flow. Pergamon; 1970. <https://doi.org/10.1016/C2009-0-06773-7>.
- [16] Chu KJ, Dukler AE. Statistical characteristics of thin, wavy films III. Structure of the large waves and their resistance to gas flow. *AIChE J* 1975;21:583–93. <https://doi.org/10.1002/aic.690210323>.
- [17] Han H, Zhu Z, Gabriel K. A study on the effect of gas flow rate on the wave characteristics in two-phase gas–liquid annular flow. *Nucl Eng Des* 2006;236:2580–8. <https://doi.org/10.1016/j.nucengdes.2006.03.015>.
- [18] Cuadros JL, Rivera Y, Berna C, Escrivá A, Muñoz-Cobo JL, Monrós-Andreu G, et al. Characterization of the gas-liquid interfacial waves in vertical upward co-current annular flows. *Nucl Eng Des* 2019;346:112–30. <https://doi.org/10.1016/j.nucengdes.2019.03.008>.
- [19] Hewitt GF, Jayanti S, Hope CB. Structure of thin liquid films in gas-liquid horizontal flow. *Int J Multiphas Flow* 1990;16:951–7. [https://doi.org/10.1016/0301-9322\(90\)90100-W](https://doi.org/10.1016/0301-9322(90)90100-W).
- [20] Schubring D, Shedd TA, Hurlburt ET. Planar laser-induced fluorescence (PLIF) measurements of liquid film thickness in annular flow. Part II: analysis and comparison to models. *Int J Multiphas Flow* 2010;36:825–35. <https://doi.org/10.1016/j.jmultiphaseflow.2010.02.002>.
- [21] Alekseenko S, Antipin V, Cherdantsev A, Kharlamov S, Markovich D. Two-wave structure of liquid film and wave interrelation in annular gas-liquid flow with and without entrainment. *Phys Fluids* 2009;21:061701. <https://doi.org/10.1063/1.3151999>.
- [22] Ishii M, Grolmes MA. Inception criteria for droplet entrainment in two-phase concurrent film flow. *AIChE J* 1975;21:308–18. <https://doi.org/10.1002/aic.690210212>.
- [23] Holowach MJ, Hochreiter LE, Cheung FB. A model for droplet entrainment in heated annular flow. *Int J Heat Fluid Flow* 2002;23:807–22. [https://doi.org/10.1016/S0142-727X\(02\)00194-7](https://doi.org/10.1016/S0142-727X(02)00194-7).
- [24] Liu L, Bai B. Generalization of droplet entrainment rate correlation for annular flow considering disturbance wave properties. *Chem Eng Sci* 2017;164:279–91. <https://doi.org/10.1016/j.ces.2017.02.023>.
- [25] Woodmansee DE, Hanratty TJ. Mechanism for the removal of droplets from a

- liquid surface by a parallel air flow. *Chem Eng Sci* 1969;24:299–307. [https://doi.org/10.1016/0009-2509\(69\)80038-2](https://doi.org/10.1016/0009-2509(69)80038-2).
- [26] Pham SH, Kawara Z, Yokomine T, Kunugi T. Detailed observations of wavy interface behaviors of annular two-phase flow on rod bundle geometry. *Int J Multiphase Flow* 2014;59:135–44. <https://doi.org/10.1016/j.ijmultiphaseflow.2013.11.004>.
- [27] Cherdantsev AV, Hann DB, Azzopardi BJ. Study of gas-sheared liquid film in horizontal rectangular duct using high-speed LIF technique: three-dimensional wavy structure and its relation to liquid entrainment. *Int J Multiphase Flow* 2014;67:52–64. <https://doi.org/10.1016/j.ijmultiphaseflow.2014.08.003>.
- [28] Andreussi P, Asali JC, Hanratty TJ. Initiation of roll waves in gas-liquid flows. *AIChE J* 1985;31:119–26. <https://doi.org/10.1002/aic.690310114>.
- [29] Alekseenko SV, Cherdantsev AV, Cherdantsev MV, Isaenkov SV, Markovich DM. Study of formation and development of disturbance waves in annular gas-liquid flow. *Int J Multiphase Flow* 2015;77:65–75. <https://doi.org/10.1016/j.ijmultiphaseflow.2015.08.007>.
- [30] Sekoguchi K, Takeishi M, Ishimatsu T. Interfacial structure in vertical upward annular flow. *J Flow Visual Soc Jpn* 1985;5:61–6. https://doi.org/10.3154/jvts1981.5.Supplement_61.
- [31] Belt RJ, Van't Westende JMC, Prasser HM, Portela LM. Time and spatially resolved measurements of interfacial waves in vertical annular flow. *Int J Multiphase Flow* 2010;36:570–87. <https://doi.org/10.1016/j.ijmultiphaseflow.2010.03.004>.
- [32] Farias PSC, Martins FJWA, Sampaio LEB, Serfaty R, Azevedo LFA. Liquid film characterization in horizontal, annular, two-phase, gas-liquid flow using time-resolved laser-induced fluorescence. *Exp Fluid* 2012;52:633–45. <https://doi.org/10.1007/s00348-011-1084-4>.
- [33] Zadrazil I, Matar OK, Markides CN. An experimental characterization of downwards gas-liquid annular flow by laser-induced fluorescence: flow regimes and film statistics. *Int J Multiphase Flow* 2014;60:87–102. <https://doi.org/10.1016/j.ijmultiphaseflow.2013.11.008>.
- [34] Cherdantsev AV, An JS, Charogiannis A, Markides CN. Simultaneous application of two laser-induced fluorescence approaches for film thickness measurements in annular gas-liquid flows. *Int J Multiphase Flow* 2019;119:237–58. <https://doi.org/10.1016/j.ijmultiphaseflow.2019.07.013>.
- [35] Isaenkov SV, Cherdantsev AV, Vozhakov IS, Cherdantsev MV, Arkhipov DG, Markovich DM. Study of primary instability of thick liquid films under strong gas shear. *Int J Multiphase Flow* 2019;111:62–81. <https://doi.org/10.1016/j.ijmultiphaseflow.2018.11.010>.
- [36] Jayanti S, Hewitt GF. Hydrodynamics and heat transfer in wavy annular gas-liquid flow: a computational fluid dynamics study. *Int J Heat Mass Tran* 1997;40:2445–60. [https://doi.org/10.1016/S0017-9310\(96\)00288-8](https://doi.org/10.1016/S0017-9310(96)00288-8).
- [37] Yun G, Ishiwatari Y, Ikejiri S, Oka Y. Numerical analysis of the onset of droplet entrainment in annular two-phase flow by hybrid method. *Ann Nucl Energy* 2010;37:230–40. <https://doi.org/10.1016/j.anucene.2009.10.019>.
- [38] Fan W, Li H, Anglart H. Numerical investigation of spatial and temporal structure of annular flow with disturbance waves. *Int J Multiphase Flow* 2019;110:256–72. <https://doi.org/10.1016/j.ijmultiphaseflow.2018.10.003>.
- [39] Saxena A, Prasser H-M. A study of two-phase annular flow using unsteady numerical computations. *Int J Multiphase Flow* 2020;126:103037. <https://doi.org/10.1016/j.ijmultiphaseflow.2019.05.003>.
- [40] Gulati S, Narayanan C, Buongiorno J, Lakehal D. Simulation of liquid entrainment in stratified and annular flow regimes using large-eddy & interface simulation. Jeju, Korea: LEIS. *Int Conf Multiph Flow*; 2013.
- [41] Sato Y, Niceno B. Large eddy simulation of upward co-current annular boiling flow using an interface tracking method. *Nucl Eng Des* 2017;321:69–81. <https://doi.org/10.1016/j.nucengdes.2017.03.003>.
- [42] Kumar P, Das AK, Mitra SK. Physical understanding of gas-liquid annular flow and its transition to dispersed droplets. *Phys Fluids* 2016;28:072101. <https://doi.org/10.1063/1.4954999>.
- [43] Yang J, Narayanan C, Lakehal D. Large Eddy & Interface Simulation (LEIS) of disturbance waves and heat transfer in annular flows. *Nucl Eng Des* 2017;321:190–8. <https://doi.org/10.1016/j.nucengdes.2016.10.054>.
- [44] Rodriguez JM. Numerical Simulation of two-phase annular flow. Rensselaer Polytechnic Institute; 2009. Doctoral dissertation.
- [45] Han H, Gabriel K. A numerical study of entrainment mechanism in axisymmetric annular gas-liquid flow. *J Fluid Eng* 2007;129:293–301. <https://doi.org/10.1115/1.2427078>.
- [46] Kim K, Choi H. Direct numerical simulation of a turbulent core-annular flow with water-lubricated high viscosity oil in a vertical pipe. *J Fluid Mech* 2018;849:419–47. <https://doi.org/10.1017/jfm.2018.408>.
- [47] Osher S, Sethian JA. Fronts propagating with curvature-dependent speed: algorithms based on Hamilton-Jacobi formulations. *J Comput Phys* 1988;79:12–49. [https://doi.org/10.1016/0021-9991\(88\)90002-2](https://doi.org/10.1016/0021-9991(88)90002-2).
- [48] Hirt CW, Nichols BD. Volume of fluid (VOF) method for the dynamics of free boundaries. *J Comput Phys* 1981;39:201–25. [https://doi.org/10.1016/0021-9991\(81\)90145-5](https://doi.org/10.1016/0021-9991(81)90145-5).
- [49] Choi H, Moin P. Grid-point requirements for large eddy simulation: chapman's estimates revisited. *Phys Fluids* 2012;24:011702. <https://doi.org/10.1063/1.3676783>.
- [50] Fan W, Anglart H. Progress in phenomenological modeling of turbulence damping around a two-phase interface. *Fluid* 2019;4:136. <https://doi.org/10.3390/fluids4030136>.
- [51] Egorov Y. Validation of CFD codes with PTS-relevant test cases. 2004.
- [52] Lakehal D. Status and future developments of Large-Eddy Simulation of turbulent multi-fluid flows (LEIS and LESS). *Int J Multiphase Flow* 2018;104:322–37. <https://doi.org/10.1016/j.ijmultiphaseflow.2018.02.018>.
- [53] Fan W, Anglart H. varRhoTurbVOF: a new set of volume of fluid solvers for turbulent isothermal multiphase flows in OpenFOAM. *Comput Phys Commun* 2020;247:106876. <https://doi.org/10.1016/j.cpc.2019.106876>.
- [54] Cherdantsev MV, Isaenkov SV, Cherdantsev AV. Shear stress in downward adiabatic annular gas-liquid flow. *J Phys: Conf Ser* 2019;1359:012026. <https://doi.org/10.1088/1742-6596/1359/1/012026>.
- [55] Alekseenko SV, Cherdantsev AV, Heinz OM, Kharlamov SM, Markovich DM. Analysis of spatial and temporal evolution of disturbance waves and ripples in annular gas-liquid flow. *Int J Multiphase Flow* 2014;67:122–34. <https://doi.org/10.1016/j.ijmultiphaseflow.2014.07.009>.
- [56] Ju Brackbill, Kothe DB, Zemach C. A continuum method for modeling surface tension. *J Comput Phys* 1992;100:335–54. [https://doi.org/10.1016/0021-9991\(92\)90240-Y](https://doi.org/10.1016/0021-9991(92)90240-Y).
- [57] Fan W, Anglart H. varRhoTurbVOF 2: modified OpenFOAM volume of fluid solvers with advanced turbulence modeling capability. *Comput Phys Commun* 2020;256:107467. <https://doi.org/10.1016/j.cpc.2020.107467>.
- [58] Fulgosi M, Lakehal D, Banerjee S, De Angelis V. Direct numerical simulation of turbulence in a sheared air-water flow with a deformable interface. *J Fluid Mech* 2003;482:319–45. <https://doi.org/10.1017/S0022112003004154>.
- [59] Ashwood AC, Vanden Hogen SJ, Rodarte MA, Kopplin CR, Rodríguez DJ, Hurlburt ET, et al. A multiphase, micro-scale PIV measurement technique for liquid film velocity measurements in annular two-phase flow. *Int J Multiphase Flow* 2015;68:27–39. <https://doi.org/10.1016/j.ijmultiphaseflow.2014.09.003>.
- [60] Vasques J, Cherdantsev A, Cherdantsev M, Isaenkov S, Hann D. Comparison of disturbance wave parameters with flow orientation in vertical annular gas-liquid flows in a small pipe. *Exp Therm Fluid Sci* 2018;97:484–501. <https://doi.org/10.1016/j.expthermflusci.2018.03.020>.
- [61] Chu KJ, Dukler AE. Statistical characteristics of thin, wavy films: Part II. Studies of the substrate and its wave structure. *AIChE J* 1974;20:695–706. <https://doi.org/10.1002/aic.690200410>.
- [62] Alekseenko S, Cherdantsev A, Heinz O, Kharlamov S, Markovich D. Analysis of spatial and temporal spectra of liquid film surface in annular gas-liquid flow. *Exp Fluid* 2013;54:1590. <https://doi.org/10.1007/s00348-013-1590-7>.
- [63] Hall Taylor N, Hewitt GF, Lacey PMC. The motion and frequency of large disturbance waves in annular two-phase flow of air-water mixtures. *Chem Eng Sci* 1963;18:537–52. [https://doi.org/10.1016/0009-2509\(63\)85014-9](https://doi.org/10.1016/0009-2509(63)85014-9).
- [64] Gill LE, Hewitt GF, Lacey PMC. Data on the upwards annular flow of air-water mixtures. *Chem Eng Sci* 1965;20:71–88. [https://doi.org/10.1016/0009-2509\(65\)85001-1](https://doi.org/10.1016/0009-2509(65)85001-1).
- [65] Wicks M, Dukler AE. Entrainment and pressure drop in concurrent gas - liquid flow: I. Air - water in horizontal flow. *AIChE J* 1960;6:463–8. <https://doi.org/10.1002/aic.690060324>.